

Using Identity Management and Policies

(LAB-M02-01)

Overview

This lab demonstrates how Identity and Access Management (IAM) evaluates permissions using identity and resource-based policies. You will learn how to create a user and the policies that can be attached to a resource, and how the evaluation of these policies results in the effective permissions. The concepts introduced in this lab act as a starting point for managing identities and resources in a single, or multiple accounts using IAM.

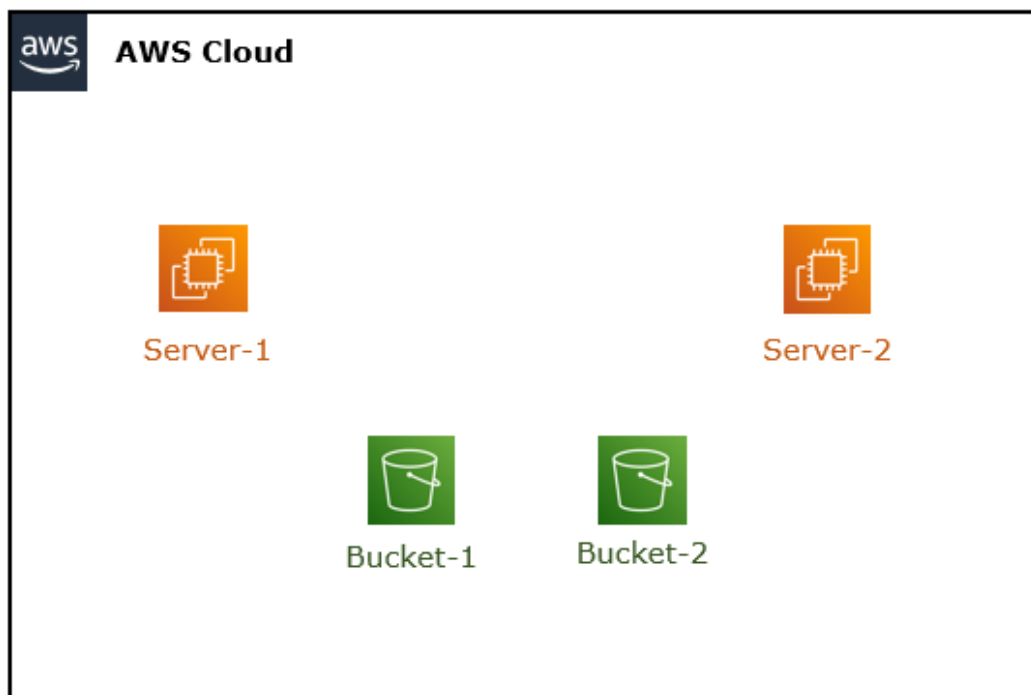
Objective

By the end of this lab, you will be able to:

- Create users
- Define permissions through policies
- Use Identity based policies to manage permissions

Task 1: Create AWS Resources

In this task, you will create EC2 instances and S3 bucket which can be managed by IAM Users.



Step 1: Choose a Region Using the Console

1. Choose the **US East (N. Virginia)** region list to the right of your account information on the navigation bar.

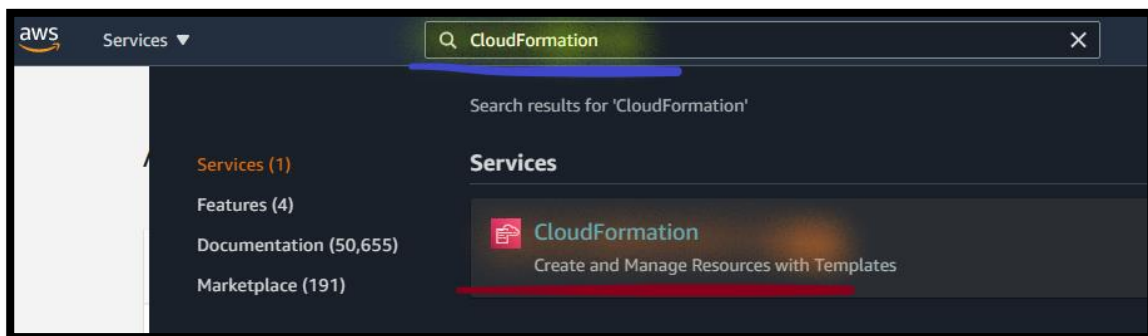
Step 2: Deploy Resources

AWS CloudFormation model and provision AWS resources in your cloud environment in automated way.

Note

- In this task, CloudFormation provision the resources.

2. In the **AWS Management Console**, on the **Services** menu Search and Select **CloudFormation**.



3. Click **Create stack** and **configure**:
 - a. In the **Create stack** page:
 - i. **Prepare template**: Select **Template is ready**.
 - ii. **Template source**: Select **Upload a template file**.
 - iii. **Choose file**: Click on **choose file** then select the **Security-LAB-Resources.yaml** file.

Note: **Security-LAB-Resources.yaml** template is provided with the lab manual.

Prerequisite - Prepare template

Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Template is ready ☐ Use a sample template ☐ Create template in Designer

Specify template
A template is a JSON or YAML file that describes your stack's resources and properties.

Template source
Selecting a template generates an Amazon S3 URL where it will be stored.

☐ Amazon S3 URL ☒ Upload a template file

Upload a template file
Choose file Security-LAB-Resources.yaml
JSON or YAML formatted file

iv. Click **Next**.

b. In the **Specify stack details** page:

i. **Stack name:** Write **Lab-Sec-01**.

Note: Leave other details as default.

Stack name

Stack name
Lab-Sec-01
Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters
Parameters are defined in your template and allow you to input custom values when you create or update a stack.

LinuxAmild
/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

ii. Click **Next**.

Note: Leave other details as default.

iii. Click **Next**.

- c. In the **Configure stack options** page:

Note: Leave other details as default.

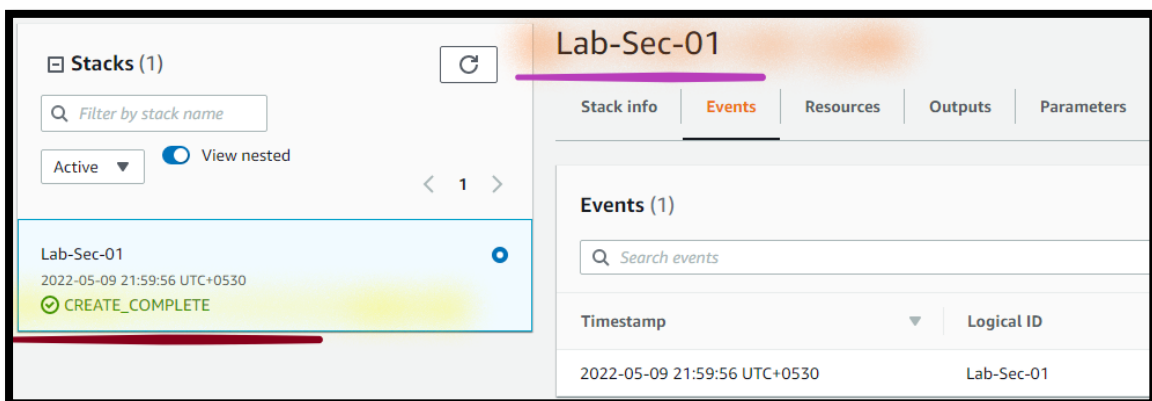
- i. Click **Next**.

- d. In the **Review Lab-Sec-01** page:

- i. Click **Create stack**.
- ii. The **stack status** is showing as **CREATE_IN_PROGRESS**.

Note: Click on **Refresh** until status is turned into **CREATE_COMPLETE**.

Note: You need to wait till you can see the **CREATE_COMPLETE** status under **Stack** on your **right site**, as shown below.

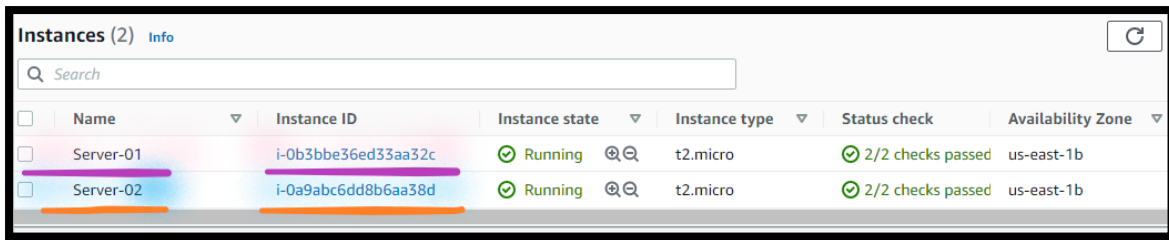


Step 2: Verify EC2 instances

4. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
- a. Select **Instances**.

Note: You can see the two servers.

- i. **Copy** the **Instance ID** of **Server-01** in the **Notepad**.



Instances (2) Info

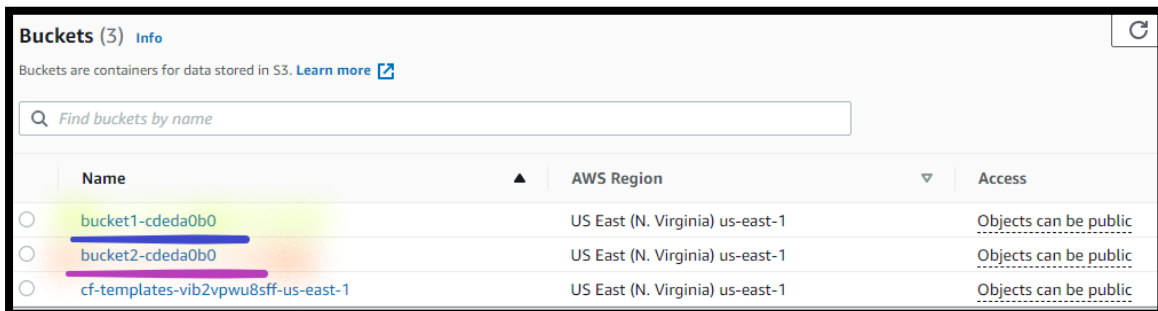
Search

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	Server-01	i-0b3bbe36ed33aa32c	Running	t2.micro	2/2 checks passed	us-east-1b
☐	Server-02	i-0a9abc6dd8b6aa38d	Running	t2.micro	2/2 checks passed	us-east-1b

Step 2: Verify S3 bucket

5. In the **AWS Management Console**, on the **Services** menu, click **S3**.
 - a. Select **Buckets**.

Note: You can see the two buckets with prefix name starting from "bucket".



Buckets (3) Info

Buckets are containers for data stored in S3. [Learn more](#)

Find buckets by name

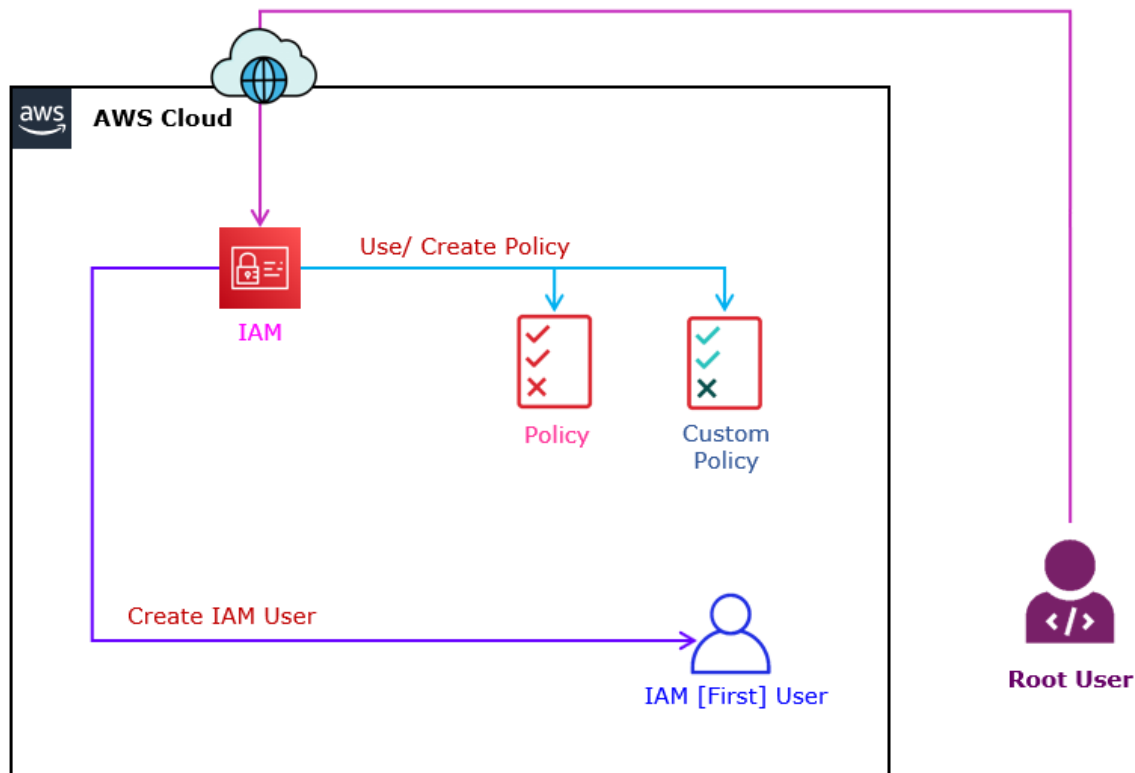
	Name	AWS Region	Access
☐	bucket1-cdeda0b0	US East (N. Virginia) us-east-1	Objects can be public
☐	bucket2-cdeda0b0	US East (N. Virginia) us-east-1	Objects can be public
☐	cf-templates-vib2vpwu8sff-us-east-1	US East (N. Virginia) us-east-1	Objects can be public

Note: Third bucket is created by CloudFormation template to store the template.

- i. **Copy** the **bucket1-xxx** name and **bucket2-xxx** in the **Notepad**.

Task 2: Create [First] IAM User [by Root User]

In this task, you will create first AWS IAM User and assign them permission to create new IAM Users. You will also restrict the first AWS IAM User, can't manage any resources except IAM services.



Step 1: Create IAM User with Permission Boundary

6. **From** the **IAM** console:
7. Select **Users**.
8. Select **Add user**.
 - a. In the **Set user details** section:
 - i. **Username**: Write **SecUser01**.

The screenshot shows the 'Set user details' section of the AWS IAM console. It includes a header 'Set user details' and a sub-header 'You can add multiple users at once with the same access type and permissions. [Learn more](#)'. Below this, there is a text input field labeled 'User name*' with the value 'SecUser01'. A blue button with a plus sign and the text 'Add another user' is located below the input field.

b. In the **Select AWS access type** section:

i. **Select AWS Access type:**

- a) Select **Access key - Programmatic access**.
- b) Select **Password - AWS Management Console access**.

Note: Leave other details as default.

Select AWS access type

Select how these users will primarily access AWS. If you choose only programmatic access, it does NOT prevent users from accessing the console using an assumed role. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Select AWS credential type* ☒ **Access key - Programmatic access**
Enables an **access key ID** and **secret access key** for the AWS API, CLI, SDK, and other development tools.

☒ **Password - AWS Management Console access**
Enables a **password** that allows users to sign-in to the AWS Management Console.

Console password* ☒ Autogenerated password
☐ Custom password

Require password reset ☒ User must create a new password at next sign-in
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

ii. Select **Next: Permissions**.

c. In the **Set permissions** page:

- i. Select **Attach existing policies directly**.
 - a) In the **Search** type **IAMFullAccess**.
 - 1) Select the **IAMFullAccess**.

▼ Set permissions

Add user to group

Copy permissions from existing user

Attach existing policies directly

Create policy

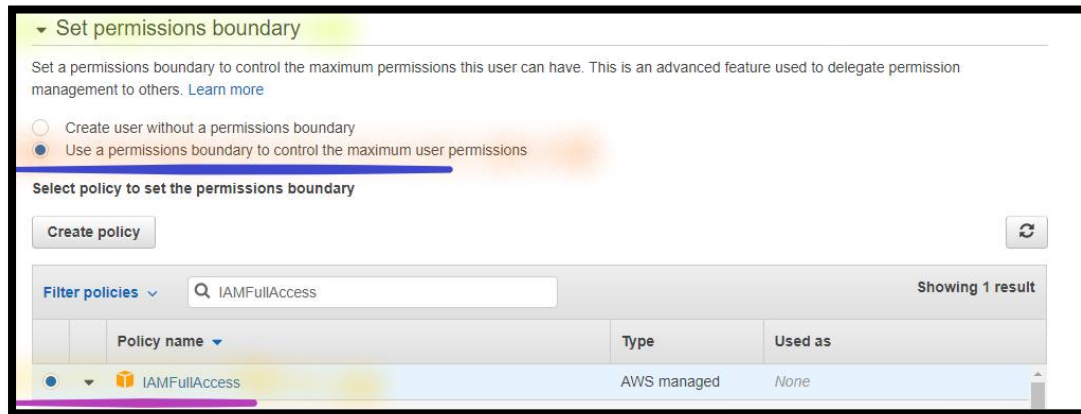
Filter policies ▼

	Policy name ▼	Type	Used as
☒	IAMFullAccess	AWS managed	None

d. In the **Set permissions boundary** page:

i. Select the **Use a permissions boundary to control the maximum user permission.**

a) In the **Search** type **IAMFullAccess**.



Info: AWS supports permissions boundaries for IAM entities (users or roles). An entity's permissions boundary allows it to perform only the actions that are allowed by both its identity-based policies and its permissions boundaries.

i. Select **Next: Tags**.

e. In the **Add tags** page:

Note: Leave other details as default.

i. Select **Next: Review**.

f. In the **Review** page:

Note: Review all the details. You can see **IAMFullAccess** under **Permssion boundry** and **IAMFullAccess** and **IAMUserChangePassword** under **Permission summary**.

User details

User name	SecUser01
AWS access type	Programmatic access and AWS Management Console access
Console password type	Autogenerated
Require password reset	Yes
Permissions boundary	 IAMFullAccess ↗

Permissions summary

The following policies will be attached to the user shown above.

Type	Name
Managed policy	IAMFullAccess
Managed policy	IAMUserChangePassword

i. Select **Create users**.

a) **Copy** the **IAM user sign-in link** in the **Notepad**.

b) Select **Download .csv** file, to download the **SecUser01 IAM user details**.

✓ **Success**
 You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: <https://504340965905.signin.aws.amazon.com/console>

[Download .csv](#)

	User	Access key ID	Secret access key	Password	Email login instructions
▶ ✓	SecUser01	AKIAWK3IQNIIUOVZA7EG ↗	***** Show	***** Show	Send email ↗

9. **Logout** from the **Root User**.

Step 2: Create Permission Boundary Policy

In this task, you will create IAM policy so that IAM user can not add/ delete permission boundary.

Update the Policy

10. **Open** the **Policy_restrict_manage_permission-boundry.txt** in the **Notepad**.

Note: Policy **Policy_restrict_manage_permission-boundry.txt** is available with the Lab manual.

- a. **Replace** the **123456789012** with your **12-digit Account Id** (*from line no. 11 and 21*)

Note: **Copy** the **12 digit Account Id** from user section.

Note: Policy performing the following tasks:

1. User can not remove the permission boundry of it's own Id.
2. User can not add new permission boundry on it's own Id.
3. User can not detach policy applied on it's own Id.
4. User can not update policy applied on it's own Id.
5. User can not delete policy applied on it's own Id.

- i. Select **File**.
- ii. Select **Save**.

Create the Policy

11. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.
12. Select **Policies**.
13. Click on **Create policy**.
 - a. Select the **JSON**.
 - i. **Remove** the **existing Policy**.
 - ii. **Copy** the **Policy_restrict_manage_permission-boundry.txt** policy, (which you have updated in the previous step).

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Deny",
7       "Action": [
8         "iam:CreatePolicyVersion",
9         "iam:DeletePolicy"
10      ],
11      "Resource": "arn:aws:iam::504340965905:policy/Policy_restrict_manage_permission-boundry"
12    },
13    {
14      "Sid": "VisualEditor1",
15      "Effect": "Deny",
16      "Action": [
17        "iam:PutUserPermissionsBoundary",
18        "iam:DeleteUserPermissionsBoundary",
19        "iam:DetachUserPolicy"
20      ],
21      "Resource": "arn:aws:iam::504340965905:user/SecUser01"
22    }
23  ]
24 }

```

- iii. Select **Next: Tags**.

Note: Leave all the details as default.

- iv. Select **Next: Review**.

- b. In the **Review Policy** page:

- i. **Name:** Write **Policy_restrict_manage_permission-boundry**.
- ii. Click on **Create Policy**.

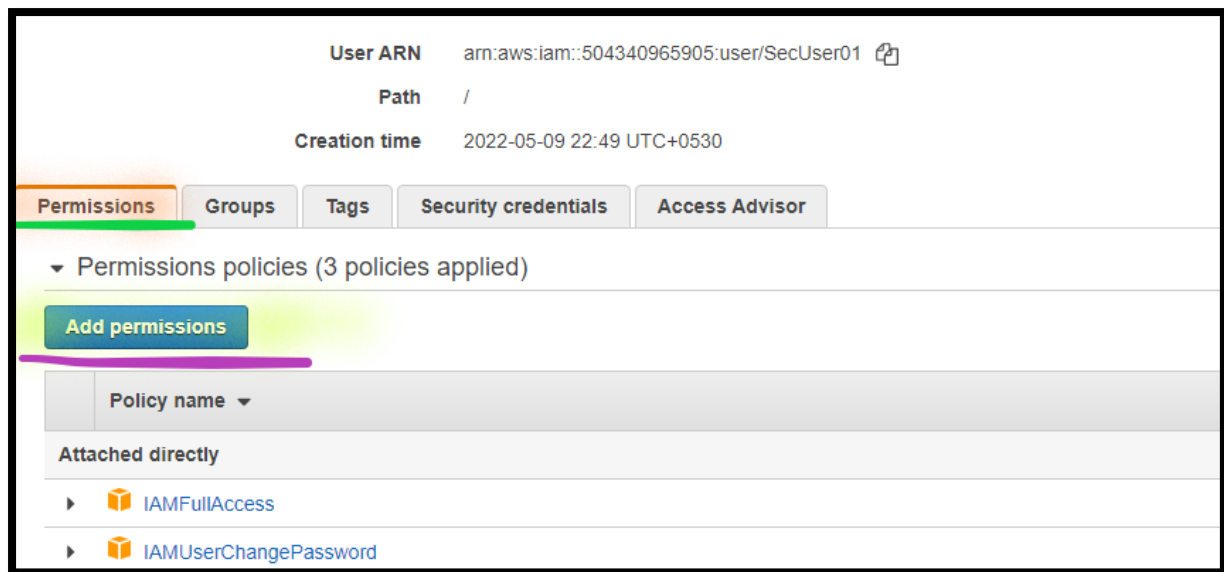
Note: You will get the message **Policy_restrict_manage_permission-boundry** been created.

Attach the Policy

- 14. **From** the **IAM** console:

- 15. Select **Users**.

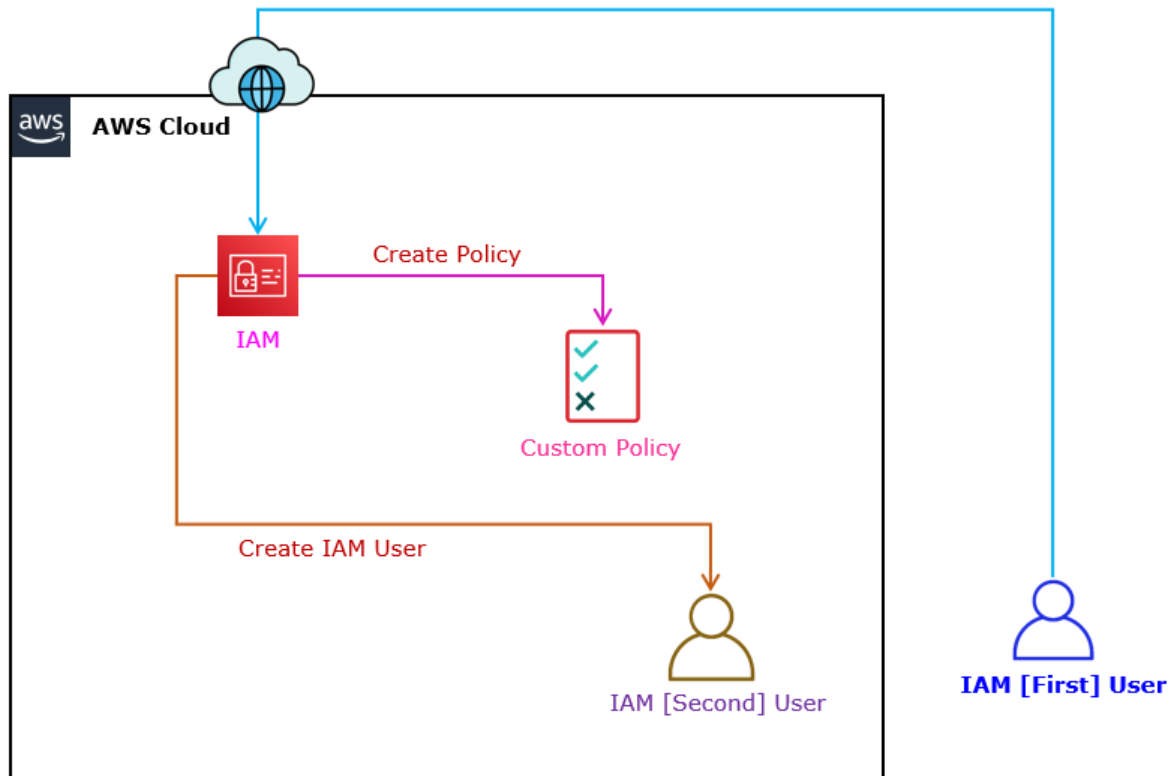
- a. Open **SecUser01**.
 - i. Select **Add permissions**.



- ii. Select **Attach existing policies directly**.
 - a) In the **Search** type **Policy_restrict_manage_permission-boundry**.
 - 1) Select the **Policy_restrict_manage_permission-boundry**.
- iii. Select **Next: Review**.
- iv. Select **Add permission**.

Task 3: Create [Second] IAM User [by First IAM User]

In this task, you will create Policies and second AWS IAM User and assign them permission to manage AWS resources (which you have created in task 1).



Step 1: Login in AWS Console

16. **Open** the **IAM user sign-in link**, from **new Browser**.

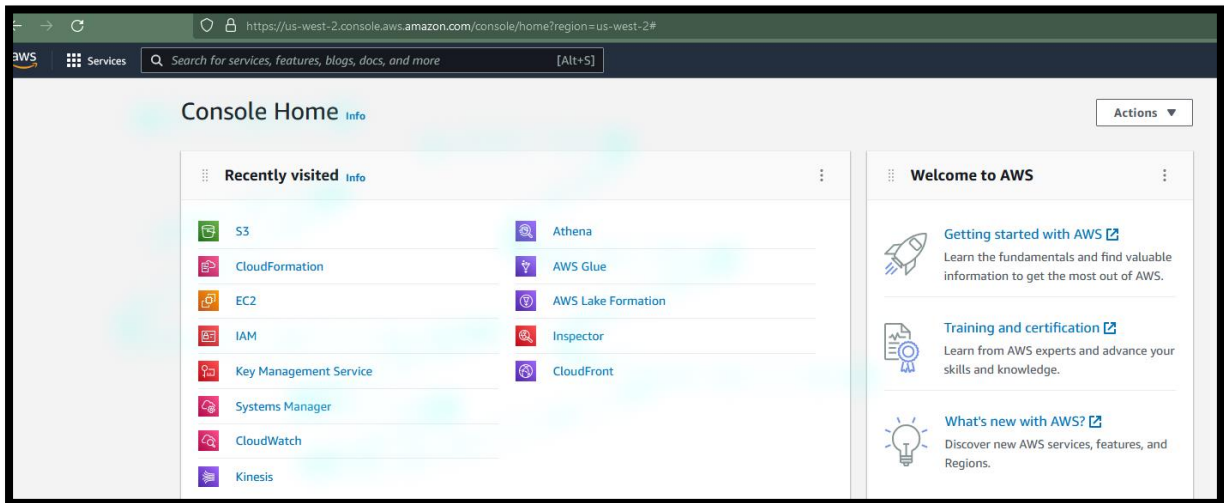
- IAM username:** Write **SecUser01**.
- Password:**

Note: **Copy** the **Password** of the IAM user **SecUser01** from **.csv file** which you have downloaded in the previous step.

- Select **Sign in**.

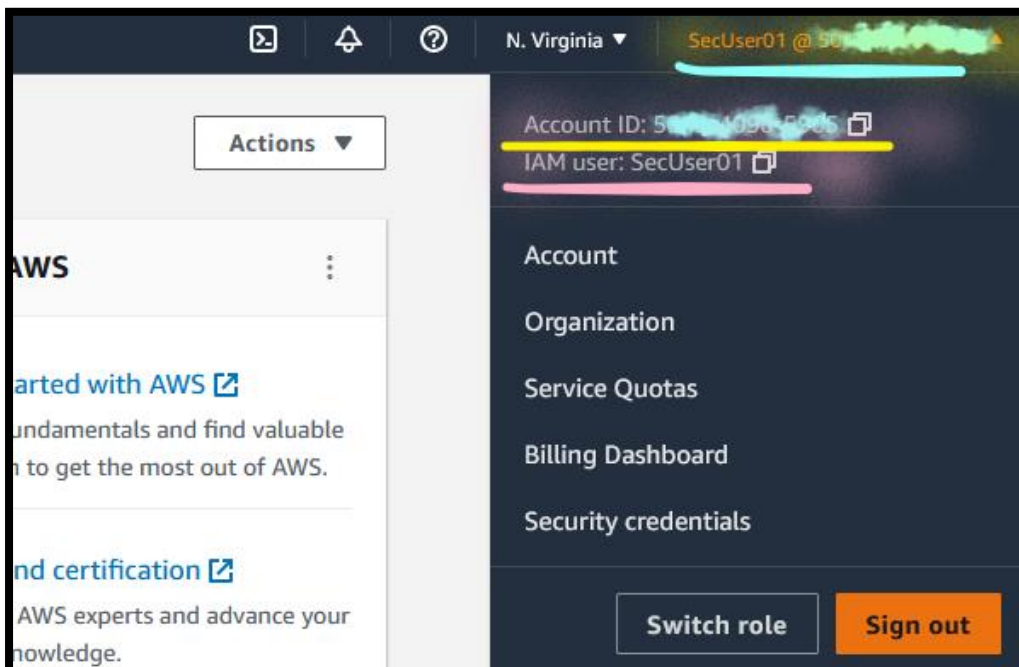
Note: When get prompt **change your password to continue**.

Note: You can see the **AWS Console**.



d. **Go to the Right**, select **IAM user name**.

Note: You can see the **IAM User details**.



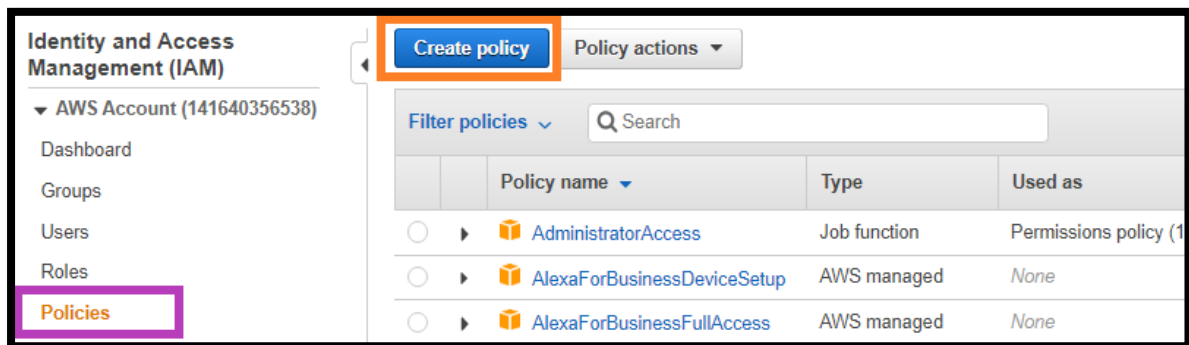
Step 2: Create S3 Policy

In this task, you will create IAM policy to manage S3 resources.

17. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.

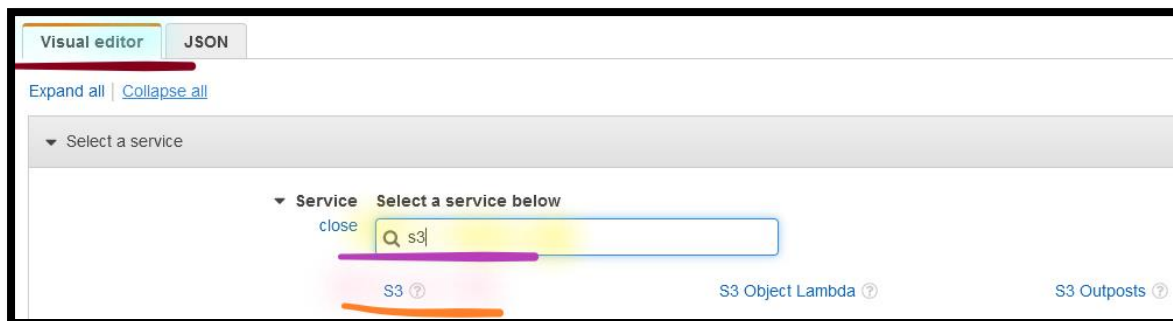
18. Select **Policies**.

19. Click on **Create policy**.



a. Select the **Visual editor**, expand the **Services**.

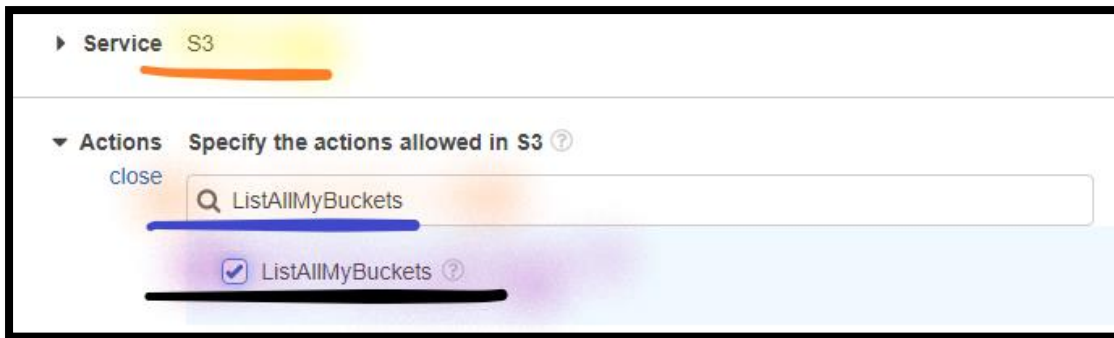
i. Search and select **S3**.



a) **Expand** the **Actions**.

1) In the **Search**, type **ListAllMyBuckets**.

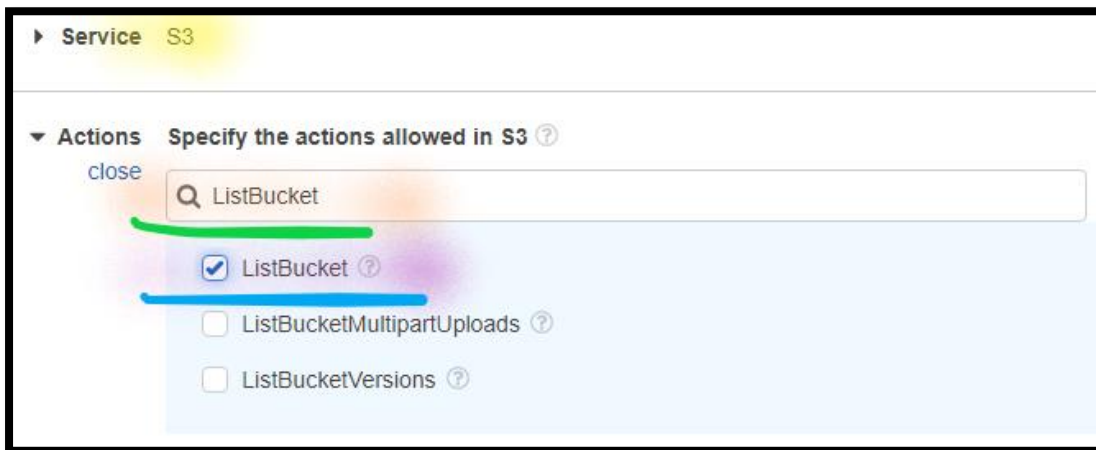
I. Select **ListAllMyBuckets**.



b) Expand the **Actions** again.

1) In the **Search**, type **ListBucket**.

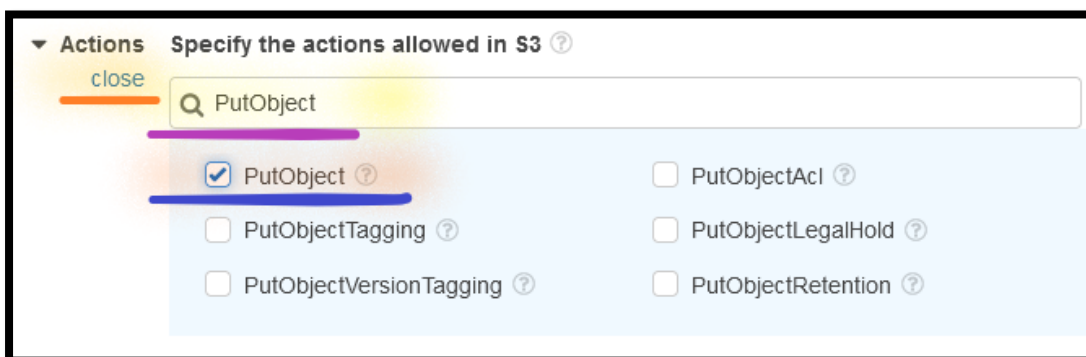
I. Select **ListBucket**.



c) Expand the **Actions** again.

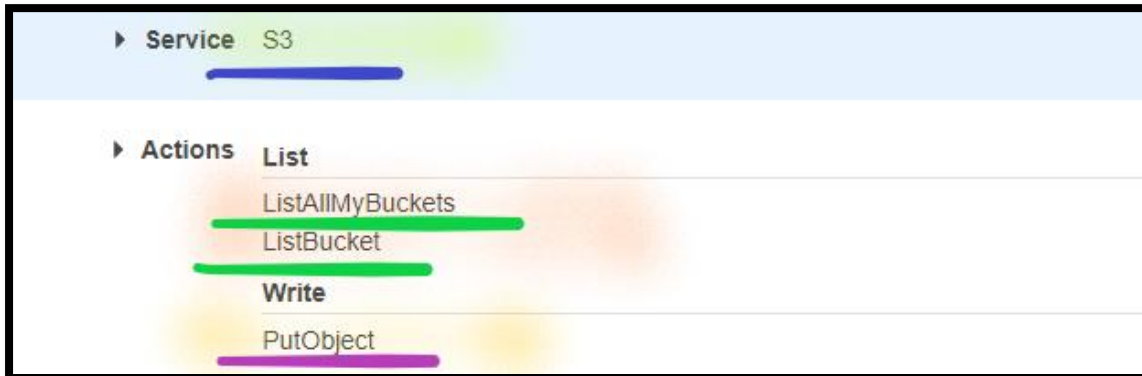
1) In the **Search**, type **PutObject**.

I. Select **PutObject**.



d) Select **Close**.

Note: You can see **1 Service** and the **3 Actions**.



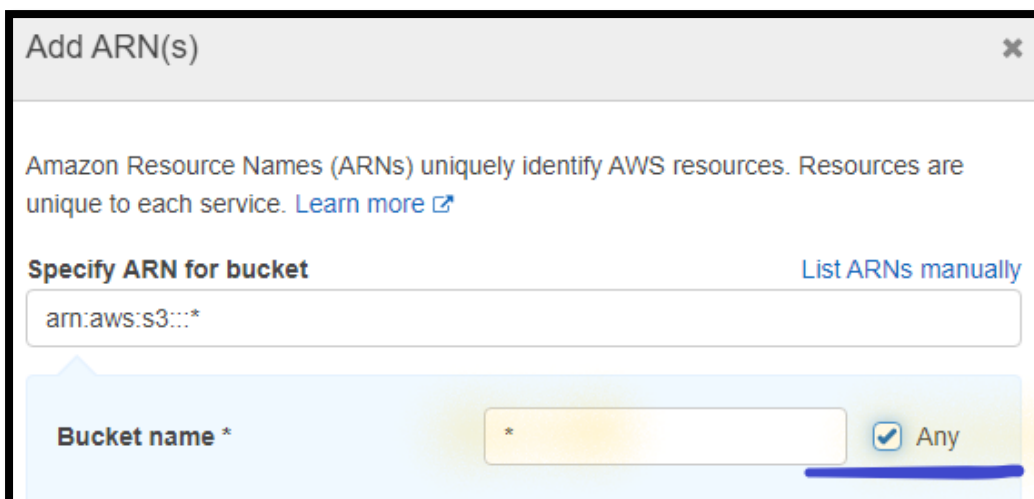
ii. **Expand the Resources.**

a) Select **Specific**.

1) Select **Add ARN** under **Bucket**.



I. **Bucket name:** Select **Any**.



II. Select **Add**.

2) Select **Add ARN** under **Object**.

i. **Bucket name:** Write **bucket1-xxx**.

Note: Replace **bucket1-xxx** with your **first bucket name**.

ii. **Object name:** Write **File1.txt**.

Note: Name is case sensitive. Ensure **F** of **File1.txt** should be **capital**.

Specify ARN for object [List ARNs manually](#)

arn:aws:s3:::bucket1-123/File1.txt

Bucket name * ☐ Any

Object name * ☐ Any

iii. Select **Add**.

3) Select **Add ARN** under **Object**.

I. **Bucket name:** Write **bucket2-xxx**.

Note: Replace **bucket1-xxx** with your **first bucket name**.

II. **Object name:** Write **File2.txt**.

Note: Name is case sensitive. Ensure **F** of **File2.txt** should be **capital**.

III. Select **Add**.

Note: You can see the **Resource details**.

▼ Resources Specific All resources

bucket Any resource of type = bucket ☒ Any

object

arn:aws:s3:::bucket1-123/file1.txt EDIT ✕

arn:aws:s3:::bucket1-123/file2.txt EDIT ✕

Add ARN to restrict access

iii. Select **Next: Tags**.

Note: Leave all the details as default.

iv. Select **Next: Review**.

v. In the **Review Policy** page:

a) **Name:** Write **Policy_s3_restricted_access**.

Name* Policy_s3_restricted_access

Use alphanumeric and '+,=, @, _' characters. Maximum 128 characters.

Description

Maximum 1000 characters. Use alphanumeric and '+,=, @, _' characters.

Summary

Filter

Service	Access level	Resource	Request condition
Allow (1 of 319 services) Show remaining 318			
S3	Limited: List, Write	Multiple	None

i. Click on **Create Policy**.

Note: You will get the message **Policy_s3_restricted_access** been created.

Step 3: Create EC2 Policy

In this task, you will create IAM policy to manage EC2 resources.

Update the EC2 Policy

20. **Open** the **Policy_ec2_restricted_access.txt** in the **Notepad**.

Note: Policy **Policy_ec2_restricted_access.txt** is available with the Lab manual.

- a. **Replace** the **i-xxx** with your **Server-01 Instance Id** (which you have copied in the previous task).

Note: EC2 Policy performing the following tasks:

1. Assign Describe EC2 instances (Listing) of All Instances.
2. Assign Start and Stop EC2 instances for Server-01.

- i. Select **File**.
- ii. Select **Save**.

Create the EC2 Policy

21. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.

22. Select **Policies**.

23. Click on **Create policy**.

- c. Select the **JSON**.
 - i. **Remove** the **existing Policy**.
 - ii. **Copy** the **Policy_ec2_restricted_access** policy, (which you have updated in the previous step).

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": "ec2:DescribeInstances",
8       "Resource": "*"
9     },
10    {
11      "Sid": "VisualEditor1",
12      "Effect": "Allow",
13      "Action": [
14        "ec2:StartInstances",
15        "ec2:StopInstances"
16      ],
17      "Resource": [
18        "arn:aws:ec2*:504340965905:instance/i-083739d4dcd452061"
19      ]
20    }
21  ]
22 }

```

- iii. Select **Next: Tags**.

Note: Leave all the details as default.

- iv. Select **Next: Review**.

d. In the **Review Policy** page:

- i. **Name:** Write **Policy_ec2_restricted_access**.
- ii. Click on **Create Policy**.

Note: You will get the message **Policy_ec2_restricted_access** been created.

Step 4: Create MFA Policy

In this task, you will create IAM policy to configure MFA.

24. **From** **IAM** console:

25. Select **Policies**.

26. Click on **Create policy**.

- a. Select the **JSON**.
 - i. **Remove** the **existing Policy**.
 - ii. **Copy** the **Policy_allow_MFA_devices** policy.

Note: Policy **Policy_allow_MFA_devices.txt** is available with the Lab manual.

Note: MFA Policy performing the following tasks:

1. Enables the MFA device and associates it with the IAM user.

- iii. Select **Next: Tags**.

Note: Leave all the details as default.

- iv. Select **Next: Review**.
- b. In the **Review Policy** page:
 - i. **Name:** Write **Policy_allows_MFA_devices**.
 - ii. Click on **Create Policy**.

Note: You will get the message **Policy_allows_MFA_devices** been created.

Step 5: Create IAM User

In this task, you will create IAM user and assign them S3, EC2 and MFA permission.

27. **From IAM** console:
28. Select **Users**.
29. Select **Add user**.
 - a. In the **Set user details** section:
 - i. **Username:** Write **AdminUser01**.

b. In the **Select AWS access type** section:

i. **Select AWS Access type:**

a) Select **Password - AWS Management Console access**.

Note: Leave other details as default.

ii. Select **Next: Permissions**.

c. In the **Set permissions** page:

i. Select **Attach existing policies directly**.

a) In the **Search** type **Policy_allows_MFA_devices**.

1) Select the **Allows_MFA_Devices**.

b) In the **Search** type **Policy_ec2_restricted_access**.

1) Select the **Policy_ec2_restricted_access**.

c) In the **Search** type **Policy_s3_restricted_access**.

1) Select the **Policy_s3_restricted_access**.

Note: Leave other details as default.

ii. Select **Next: Tags**.

d. In the **Add tags** page:

Note: Leave other details as default.

i. Select **Next: Review**.

e. In the **Review** page:

Note: Review all the details.

- i. Select **Create users**.
 - a) **Copy** the **IAM user sign-in link** in the **Notepad**.
 - b) Select **Download .csv** file, to download the **AdminUser01 IAM user details**.

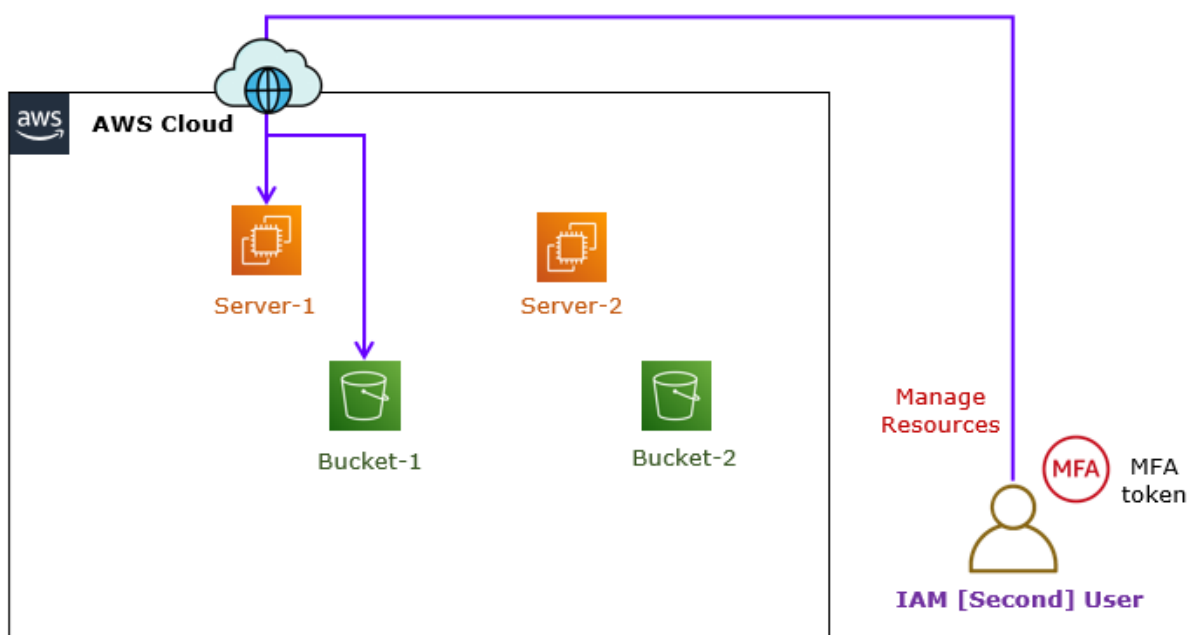
Step 6: Log out

30. **Go to the Right**, select **IAM username**.

- a. Select **Sign Out**.

Task 4: Manage Resources by [Second IAM User]

In this task, you will test the second IAM User to manage the AWS S3 bucket and EC2 instances. You will also configure the MFA by second IAM User



Step 1: Login in AWS Console

31. **Open** the **IAM user sign-in link**.
- a. **IAM username**: Write **AdminUser01**.
 - b. **Password**:

Note: Copy the Password of the IAM user AdminUser01 from .csv file which you have downloaded in the previous step.

- c. Select **Sign in**.

Note: When get prompt change your password to continue.

Note: You can see the AWS Console.

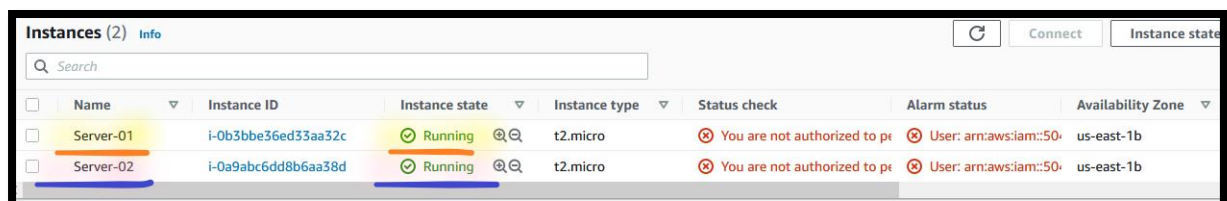
Step 2: Manage AWS EC2 Instance

32. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

33. Dropdown and select **US East (N. Virginia)** region.

- a. Select **Instances**.

Note: You can see the two instances as **Running**.



The screenshot shows the AWS Management Console 'Instances' page. It displays a table with two instances, both in a 'Running' state. The table columns include Name, Instance ID, Instance state, Instance type, Status check, Alarm status, and Availability Zone. The first instance is 'Server-01' with ID 'i-0b3bbe36ed33aa32c' and the second is 'Server-02' with ID 'i-0a9abc6dd8b6aa38d'. Both are t2.micro instances in the us-east-1b availability zone. The status check for both shows a red 'X' with the message 'You are not authorized to perform this action. Please contact your administrator.'

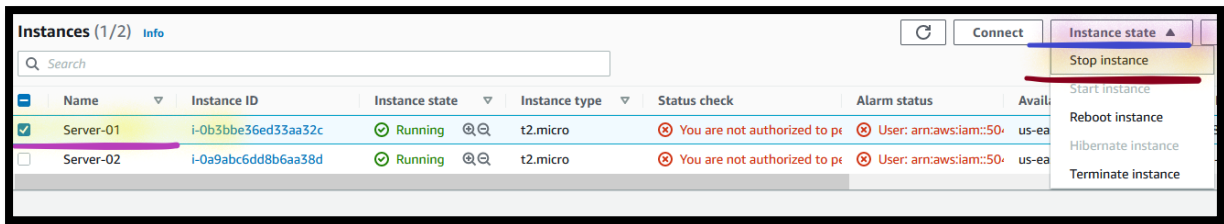
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
Server-01	i-0b3bbe36ed33aa32c	Running	t2.micro	You are not authorized to perform this action. Please contact your administrator.	User: arn:aws:iam::50...	us-east-1b
Server-02	i-0a9abc6dd8b6aa38d	Running	t2.micro	You are not authorized to perform this action. Please contact your administrator.	User: arn:aws:iam::50...	us-east-1b

Stop the First Instance

34. Select the **First Instances**.

- a. Select **Instance state**.
- b. Select **Stop instance**.

Note: Observe the **Output**.



Stop the Second Instance

35. Select the **Second Instances**.

- Select **Instance state**.
- Select **Stop instance**.

Note: Observe the **Output**.

Step 3: Manage AWS S3

36. In the **AWS Management Console**, on the **Services** menu, click **S3**.

37. Select **Buckets**.

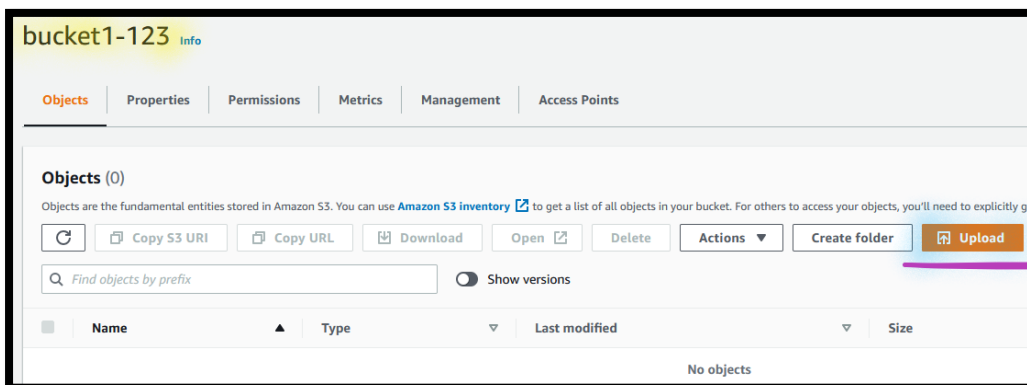
Note: You can see the **buckets**.

Upload the Object in First Bucket

- Open the **bucket1-123**.

Upload the First Object

- Select **Upload**.



- ii. Select **Add file**.
 - a) **Navigate** and select the **File1.txt**.

Note: **File1.txt** is available with the lab manual.

- b) Select **Upload**.

Note: Observe the **Output**.

- c) Select **Close**.

Upload the Second Object

- b. **From** the **bucket1-123** console:
 - i. Select **Upload**.
 - ii. Select **Add file**.
 - a) **Navigate** and select the **File2.txt**.

Note: **File2.txt** is available with the lab manual.

- b) Select **Upload**.

Note: Observe the **Output**.

- c) Select **Close**.

Upload the Third Object

- c. **From** the **bucket1-123** console:
 - i. Select **Upload**.
 - ii. Select **Add file**.

- a) **Navigate** and select the **File3.txt**.

Note: **File3.txt** is available with the lab manual.

- b) Select **Upload**.

Note: Observe the **Output**.

- c) Select **Close**.

Upload the Object in Second Bucket

38. **From** the **S3** console:

- a. Select **Buckets**.
- b. Open the **bucket2-123**.

Upload the Object

- i. Select **Upload**.
- ii. Select **Add file**.
 - a) **Navigate** and select the **File1.txt**.

Note: **File1.txt** is available with the lab manual.

- b) Select **Upload**.

Note: Observe the **Output**.

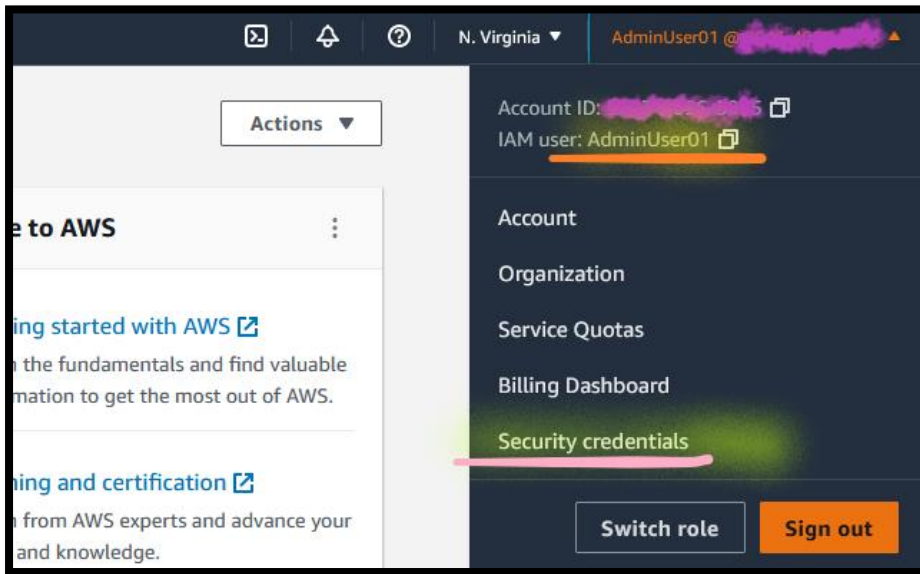
- c) Select **Close**.

Step 4: Configure the MFA

39. Install **Google Authenticator** on your mobile.

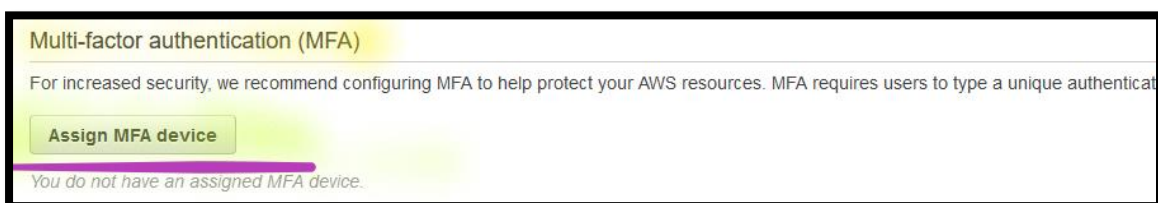
40. From the **AdminUser01** IAM User **console**:

- a. **Go to the Right**, select **IAM username**.
- b. Select **Security credentials**.



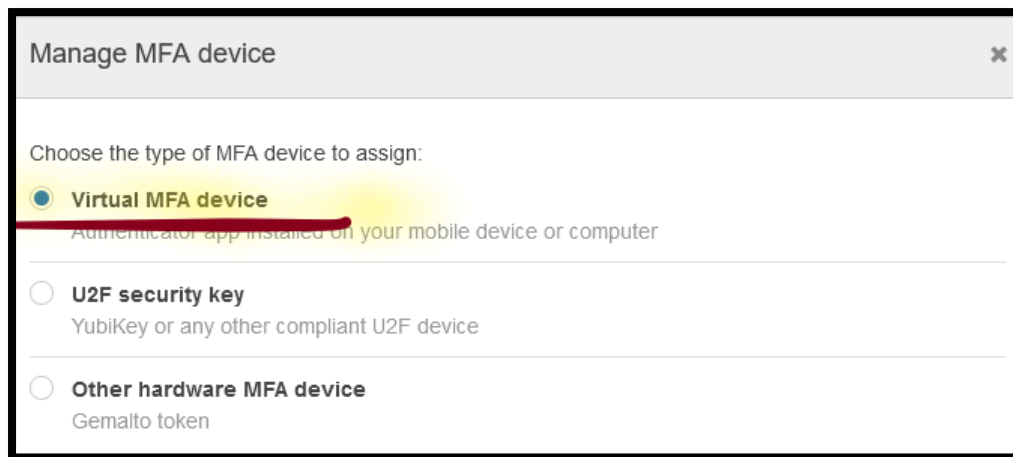
c. From the **My security credentials** page:

- i. Select **Assign MFA Device**.



ii. In the **Manage MFA Device** section:

- a) Select **Virtual MFA Device**.



b) Select **Continue**.

iii. In the **Set up virtual MFA device** section:

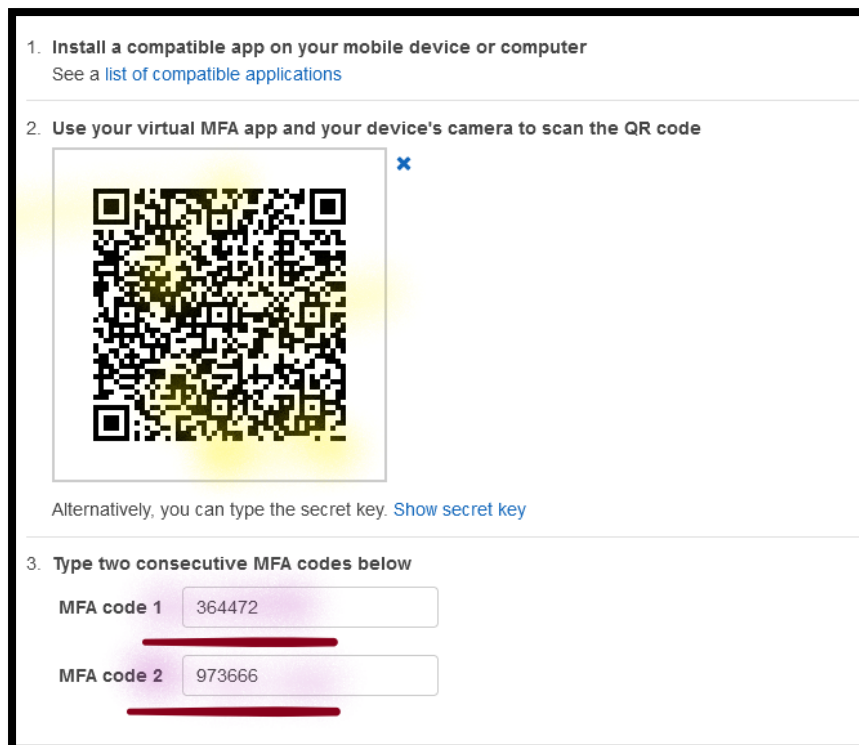
a) Select **Show QR Code**.



b) Open the **Google Authenticator App** (from your mobile) and **Scan the QR code**.

1) Provide the **First Code**.

2) Provide the **Second Code**.



1. Install a compatible app on your mobile device or computer
See a [list of compatible applications](#)

2. Use your virtual MFA app and your device's camera to scan the QR code



Alternatively, you can type the secret key. [Show secret key](#)

3. Type two consecutive MFA codes below

MFA code 1

MFA code 2

3) Select **Assign MFA**.

c) Select **Close**.



Set up virtual MFA device

✓ You have successfully assigned virtual MFA
This virtual MFA will be required during sign-in.

Close

Step 12: Log out

41. **Go to the Right**, select **IAM username**.

a. Select **Sign Out**.

Step 13: Login in AWS Console

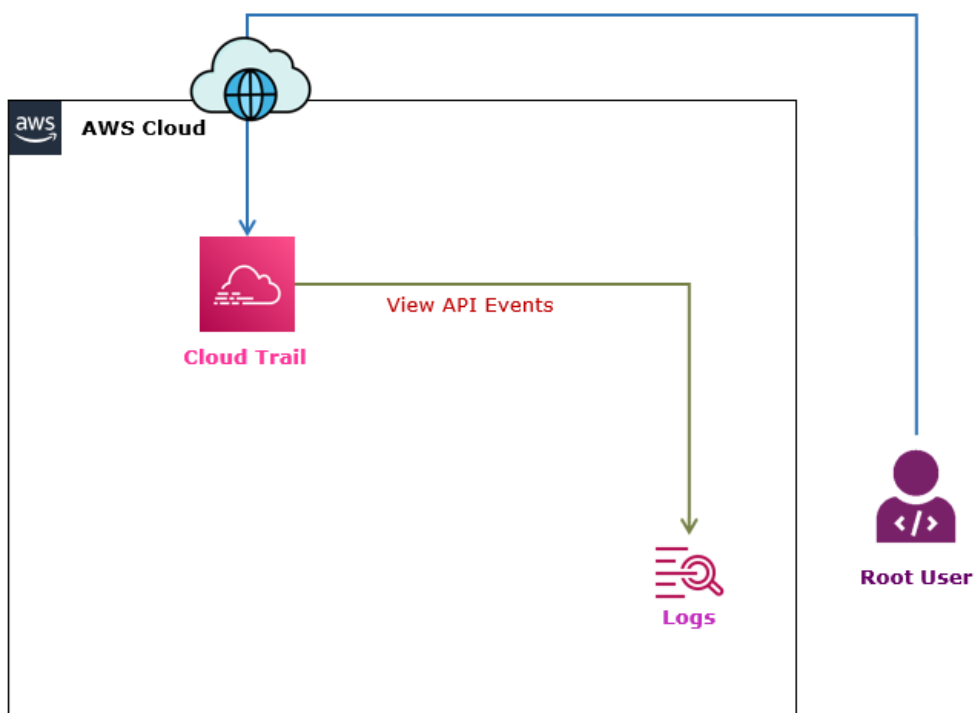
42. **Open** the **IAM user sign-in link**, from **new Browser**.
- IAM username:** Write **AdminUser01**.
 - Password:** Type your **Password** (which you have changed in the previous step).
 - MFA Code:** Provide your **MFA code** (from your mobile).
 - Select **Sign in**.

Note: You can see the **AWS Console**.

43. **Go to the Right**, select **IAM username**.
- Select **Sign Out**.

Task 5: View API Calls by [Root User]

In this task, you will view the API calls. You will also configure and store the trails into S3 bucket for compliance purpose and integrate the trail with CloudWatch to generate the alerts.



Step 1: View API Calls

44. **Login** into the **AWS Management Console** using **Root User**.
45. In the **AWS Management Console**, on the **Services** menu, click **CloudTrail**.
46. Dropdown and select **US East (N. Virginia)** region.

View Events based on Event name

47. Click **Event History**.
 - a. **First Section**: Dropdown and select **Event name**.
 - b. **Second Section**: Write **CreateUser**.

Note: CreateUser will display the events related to IAM User Creation.

	Event name	Event time	User name	Event source
☐	CreateUser	March 22, 2022, 12:03:24 (UTC+05:30)	SecUser01	iam.amazonaws.com
☐	CreateUser	March 21, 2022, 22:43:55 (UTC+05:30)	root	iam.amazonaws.com

- i. Open the First **event** created by **Root**.

Note: You can see the Event **details**.

CreateUser Info		
Details Info		
Event time	AWS access key	AWS region
March 21, 2022, 12:00:55 (UTC+05:30)	ASIAXK3IQNIIZQQ3QIZB	us-east-1
User name	Source IP address	Error code
root	AWS Internal	-
Event name	Event ID	Read-only
CreateUser	f5171ddb-b020-4a90-9578-f029dd0fa10f	false
Event source	Request ID	
iam.amazonaws.com	29bda21b-fae2-4a99-be48-0384fbfdde37	

Note: You can see the **Event record**.

```
Event record Info
{
  "eventVersion": "1.08",
  "userIdentity": {
    "type": "Root",
    "principalId": "AIDAXK3IQNIIZQQ3QIZB",
    "arn": "arn:aws:iam::504340965905:root",
    "accountId": "504340965905",
    "accessKeyId": "ASIAXK3IQNIIZQQ3QIZB",
    "sessionContext": {
      "sessionIssuer": {},
      "webIdFederationData": {},
      "attributes": {
        "creationDate": "2022-03-21T05:04:29Z",
        "mfaAuthenticated": "false"
      }
    }
  },
  "eventTime": "2022-03-21T06:30:55Z",
  "eventSource": "iam.amazonaws.com",
  "eventName": "CreateUser",
  "awsRegion": "us-east-1",
  "sourceIPAddress": "AWS Internal",
  "userAgent": "AWS Internal",
  "requestParameters": {
    "userName": "SecUser01",
    "tags": [],
    "permissionsBoundary": "arn:aws:iam::aws:policy/IAMFullAccess"
  },
}
```

```
"responseElements": {
  "user": {
    "path": "/",
    "userName": "SecUser01",
    "userId": "AIDAXK3IQNIIZQQ3QIZB",
    "arn": "arn:aws:iam::504340965905:user/SecUser01",
    "createDate": "Mar 21, 2022 6:30:55 AM",
    "permissionsBoundary": {
      "permissionsBoundaryType": "Policy",
      "permissionsBoundaryArn": "arn:aws:iam::aws:policy/IAMFullAccess"
    }
  }
},
"requestID": "29bda21b-fae2-4a99-be48-0384fbfdde37",
"eventID": "f5171ddb-b020-4a90-9578-f029dd0fa10f",
"readOnly": false,
"eventType": "AwsApiCall",
"managementEvent": true,
"recipientAccountId": "504340965905",
"eventCategory": "Management",
"sessionCredentialFromConsole": "true"
```

View Events based on Username

48. From the **Event History**.

- First Section:** Dropdown and select **User name**.
- Second Section:** Write **SecUser01**.

Note: You can see all the events created by IAM User.

Event history (50+) Info

Event history shows you the last 90 days of management events.

User name: SecUser01

30m 1h 3h 12h Clear

☐	Event name	Event time	User name	Event source	Resource type
☐	AttachUserPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::Policy
☐	AttachUserPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::Policy
☐	GetPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	-
☐	AttachUserPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::Policy
☐	GetPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	-
☐	GetPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	-
☐	AttachUserPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::Policy
☐	GetPolicy	March 22, 2022, 12:03:25 (UTC+05:30)	SecUser01	iam.amazonaws.com	-
☐	ListUsers	March 22, 2022, 12:03:24 (UTC+05:30)	SecUser01	iam.amazonaws.com	-
☐	CreateUser	March 22, 2022, 12:03:24 (UTC+05:30)	SecUser01	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::User, AWS::IAM::Policy

Challenge:
Task
[If time permits]

- Attempt to test the permission boundry.

Note: Refer the **Challenge task.pdf manual** to test the permission boundry, in case you are facing issues to perform the **Task**.

Optional:
Task
[If time permits]

- Refer the **Optional task.pdf manual** to Store Cloud Trail events in the S3 bucket.

Task 7: Cleanup the Environment by [Root User]

Step 1: Empty the S3 Bucket

49. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

- a. Select **bucket1-123** bucket.
 - i. Select **Empty**.
 - ii. Type **permanently delete** to *delete all the objects*.
 - iii. Select **Empty**.
 - iv. Select **Exit**.

Step 2: Delete EC2 Instances and S3 Buckets

50. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.

- a. Select **Stack**.
- b. Select **Lab-Sec-01**.
 - i. Select **Delete**.
 - ii. Select **Delete stack**.